MILITARY SPECIFICATION

BALLISTICS COMPUTER SUBASSEMBLY: 8635276 (INPUT UNIT)

1. SCOPE

1.1 This specification covers one type of electrical-mechanical assembly known as Ballistics Computer Subassembly: 8635276, hereinafter referenced as 'Input Unit" (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Material: General Specification Governing the Manufacture and Inspection of.
MIL-P-14232	-	Parts Equipment and Tools for Ordnance Material: Packaging of.
MIL-I-45208	-	Inspection System Requirements.,
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	-	Quality Assurance Terms and Definitions.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

F8635276	-	Ballistics Computer Subassembly: 8635276 (Input Unit).
1EL 8635276	-	List of Inspection Documents for Ballistics Computer Subassembly Computer Subassembly: 8635276 (Input Unit).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication. The input unit shall be manufactured in accordance with Drawing F8635276 and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence to insure inherent quality and performance capabilities.

3.3 General specification. The contractor shall be responsible for adherence to and compliance with the following requirements of Specification MIL-F-13826:

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Part identification and marking.
(e) Workmanship.

3.4 Environmental

3.4.1 *Operating tempratures.* The input unit shall meet all the requirements of 3.6 and 3.7 inclusive while exposed and thermally stabilized at ambient temperatures of -40° and + 150° Fehrenheit (F). Upon return to standard ambient temperature, the input unit shall meet the requirements of 3.4.2.

3.4.2 *Vibration.* The input unit shall meet the requirements of 3.5 to 3.7 inclusive after being vibrated in a vertical plane for five minutes ± 15 seconds at an ampltude of 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second (cps).

3.4.3 *Water resistance (sealing).* The interior of the input unit shall show no evidence of water penetration or other damage after being subjected to a simulated rain fall for a time duration of 15 minutes ± 15 seconds.

3.5 Electrical

3.5.1 *Continuity.* The continuity of the input unit shall conform to the Electrical Schematic Drawing D8635307.

3.5.2 *Dielectric strength.* There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength when the insulated areas of the input unit are subjected to an alternating current (ac) root mean square (rms) potential of 300 volts plus twice the rated voltage at a frequency of 60 cps for a period of not more than 30 seconds not less than 5 seconds.

3.5.3 *Insulation resistance.* All insulation between ungrounded circuits and ground, and between isolated circuits, shall have a minimum resistance of 10 megohms with 500 volts dc applied.

3.6 Performance.

3.6.1 *Torque.* The running torque required to rotate the input shaft shall not exceed 10

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ounce-inches when checked at standard ambient temperatures. The value obtained at standard ambient temperature shall not be exceeded by more than 5 ounce-inches when checked at extreme temperatures of 3.4.1.

3.6.2 *Range data.* The range dial of the input unit shall indicate ranges within the tolerance of plus or minus 10 meters in accordance with Table I when the input shaft is positioned on each check point of table I. The input shaft shall indicate 356.6 degrees from zero position counterclockwise when the 1200 meter index on the range dial and the scribe lines of the input shaft and housing are in coincidence.

3.6.3 *Output voltages resistor R201.* The output voltage ratios of resistor R201 shall agree with the ratio settings listed in table I for input shaft positions and tolerances shown. There shall be no discontinuities in voltage output greater than 3 steps of resistor resolution between 450 and 4700 meters.

3.6.4 *Output voltages resistor R202.* The output voltage ratios for sections 1,2 and 3 of resistor R202 shall agree with the ratio settings listed in table I for input shaft positions and tolerances shown. There shall be no discontinuities in voltage output greater than 3 steps of resistor resolution between 450 and 4700 meters.

3.7 Illumination. The range dial when illuminated from a 24 volt power source shall be legible when observed in a darkened area.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or purchase order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *General provisions.* The quality assurance provisions of this specification and of other documents referenced herein shall form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL-I-45208 and the component inspection requirements of Specification MIL-F-13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL-STD-109.

TABLE I

Input shaft position (ccw)		Range in meters (infor-	Voltage ratio settings				Voltage ratio-tolerances	
Turns	Degrees	mational)	R201	R202-1	R202-2	R202-3	R201	R202-1, 2, 3
0	0.0	450	.....	.029234	.300250	.021968	± .005000	± .005000
4	180.2	700	.058823	.....	.416337	.....		
8	48.1	1000	.129411	.072613	.....	.....		
9	356.6	1200	.....	.....	.500000	.079967		
11	201.9	1400	.223529	.110683	.....	.....	± .005000	± .005000

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TABLE I (Cont'd)

Input shaft position (ccw)		Range in meters (infor-	Voltage ratio settings				Voltage Ratio-tolerances	
Turns	Degrees	mational)	R201	R202-1	R202-2	R202-3	R201	R202-1, 2, 3
13	193.8	1700	.294117	.....	.....	.....	± .005000	± .005000
14	43.4	1800	.....	.155751	.....	.....		
15	69.7	2000	.364705	.....	.....	.....		
16	59.2	2200	.....	.....	.....	.208402		
16	222.2	2300	.....	.225337	.....	.....		
16	301.1		.447058	.....	.....	.....		
17	18.3	2400	.....	.....	.558060	.....		
18	21.7		.517647	.....	.....	.....		
18	223.6	2800	.....	.315883	.....	.....		
18	352.2	2900	.576470	.....	.....	.....		
19	116.5	3000	.....	.359903	.....	.....		
19	353.2	3200	.647058	.....	.....	.....		
20	215.5	3400	.....	.....	.....	.464717		
20	321.8	3500	.717647	.....	.....	.....		
21	214.8		.776470	.....	.....	.....		
21	358.6	3900	.....	.638376	.....	.....		
22	91.5	4000	.835294	.....	.....	.....		
23	81.0	4400	.929411	.....	.....	.....		
23	322.8	4700	.....	.....	.587312	.....	± .005000	± .005000

4.2 Inpsection provisions.

4.2.1 *Inspection lot.* Unless otherwise specified by the contracting officer, inspection lot size, formation and presentation of lots shall be in accordance with "Submission of Products" as specified in Standard MIL-STD-105.

4.2.2 *Examination and tests.* Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL-F-13926 and the Supplementary Quality Assurance Provisions, listed on IEL 8635276. Examinations and tests related to section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL-STD-105, and the inspection level and sampling plan specified in table II titled, "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with Specification MIL-P-14232 and section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual nonconforming items.

TABLE II. *Classification of defects*

Use Level 11 of Table I with Sampling Plan Table 11-A Standard MIL-STD-105

	Requirement	*Test procedure*
Critical: None.		
Major: AQL 1.0%.		
101. Physical failure, post vibration	3.4.2	4.5.1.2
102. Continuity	3.5.1	4.5.2

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103. Torque	3.6.1	4.5.5
104. Range Data	3.6.2	4.5.6
105. Output voltage, R201	3.6.3	4.5.7
106. Output voltage, R202 (sec. 1, 2 & 3)	3.6.4	4.5.8
107. Illumination	3.7	4.5.9

Minor: None.

Note. The above tests shall be performed at standard ambient temperature (60° to 90°F.).

4.2.3 *Disposition of nonconforming product.* Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in Standard MIL-STD-105.

4.3 Control tests.

4.3.1 *General.* One input unit shall be selected at random from each one hundred produced, or from each month's production, whichever occurs first. The control sample shall meet the following requirements and tests:

Control test	*Requirement*	*Test procedure*
301 Materials	3.1	as per spec. or dwg.
302 Fabrication	3.2	visual-per dwg.
303 General Specification	3.3	visual-MIL-F-13926

4.3.2 *Environmental and electrical.* Three input units shall be selected at random from each fifty produced, or from each month's production, whichever occurs first. The control sample shall meet the following requirements and tests:

Control test	*Requirement*	*Test procedure*
304 Operating temperature	3.4.1	4.5.1.1
305 Water resistance	3.4.3	4.5.1.3
306 Dielectric strength	3.5.2	4.5.3
307 Insulation resistance	3.5.3	4.5.4

4.3.3 *Control test failure.* Should any one item of the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4. Inspection equipment. Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Measuring and Test Equipment" requirement of Specification MIL-I-45208 and all applicable requirements specified in Specification MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 *Contractor design.* The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL-F-13926. Concept, construction, materials, dimensions and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the

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prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2 *Inspection equipment accuracy requirements.* The devices required in the test methods and procedures herein, to measure the electrical values specified in section 3 shall provide reading to the accuracy specified below:

Item number	*Description*
1. Ratio Standard	Ratio range from .00000 to 1.00000 accuracy to ± .01%. Input impedance at 400 cps, 10,000 ohms minimum. Series impedance - 10 ohms maximum. Phase shift 0.01° maximum.
2. Oscilloscope	Minimum sensitivity of 20 millivolts, full scale, vertical channel. External synchronization. Input impedance - 1 megohm ± 10%.
3. A.C. Power Supply	0 - 24 volts rms, 400 cps ± 20 cps, 5% maximum harmonic distortion.
4. D.C. Power Supply	24 volts ± 5%, maximum ripple - 1.0 volt peak to peak, 300 milliampere output.

4.5 Test methods and procedures.

4.5.1 *Environmental.*

4.5.1.1 *Operating temperatures.* This test shall be conducted in conformance with the temperatures specified in 3.4.1. The test equipment shall be in accordance with the "Test Facilities" requirements specified in Specification MIL-F-13926. While thermally stabilized at the specified temperatures, the input unit shall be subjected to the tests specified in 4.5.5 to 4.5.9 inclusive to determine conformance with the requirements referenced therein.

4.5.1.2 *Vibration.* To determine conformance to 3.4.2 the input unit shall be mounted in its normal position using a suitable mounting apparatus to attach it to a vibrating machine or fixture, with provisions for securing universal assembly. Test equipment shall be in accordance with the "Test Facilities" portion of Specification MIL-F-13926 and shall be capable of providing vibration at the amplitude and frequency for the time specified in 3.4.2. After vibration, the input unit shall be given a visual tactile examination for damage and loose parts. The input unit shall then be subjected to the tests specified in 4.5.2 to 4.5.9 inclusive to determine conformance with the respective requirements referenced therein.

4.5.1.3 *Water resistance.* To determine conformance to 3.4.3 this test shall be conducted in accordance with Procedure 1 of Specification MIL-F-13926 except for the time duration specified in 3.4.3. The input unit shall show no evidence of water penetration as a result of this test when visually examined.

4.5.2 *Continuity.* The input unit shall be tested, using standard measuring equipment, to demonstrate continuity of the circuits in accordance with Schematic Drawing D8635307 to determine compliance with 3.5.1.

4.5.3 *Dielectric strength.* The input unit shall be tested in accordance with Method 301 of Standard MIL-STD-202 and shal meet the requirements specified in 3.5.2.

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4.5.4 *Insulation Resistance.* Immediately after satisfactorily withstanding the test for dielectric strength, the input unit shall be given the insulation resistance test in accordance with Method 302, test condition B, of Standard MIL-STD-202, and shall meet the requirements specified in 3.5.3.

4.5.5 *Torque.* This test shall be performed with the aid of special testing devices affixed to the input shaft. When the input shaft is rotated in a clockwise and counterclockwise direction, the measured torque shall not exceed the value specified in 3.6.1 at the temperatures indicated in 3.4.1.

4.5.6 *Range data.* To determine conformance to 3.6.2, attach an angle measuring device and a turn counting device to the rangefinder input shaft. This device shall be capable of measuring the angles of table I to an accuracy of 0.1 of a degree. A magnifying lens of at least 4 power shall be provided to ascertain the alignment of scribe marks on the input shaft and housing. With these scribe lines aligned, 1200 meters registering on the range dial, and the turn counter-angle device recording 9 turns and 356.6 degrees, the unit shall be considered ready for a range dial check. By means of the hand crank, return the range scale dial to the 450 position and the counter-angle device to zero. Proceed to check each position of the input shaft as shown on table I.

4.5.7 *Output voltages resistor R201.* To determine conformance to 3.6.3, the device used in 4.5.6 shall be attached to the input shaft. An ac ratio standard (4.4.2) shall be connected in parallel with resistor R201, by connection at pins S and X of J201. Connect a high impedance oscilloscope (4.4.2) between pin small j of J201 and the ratio standard slider. Apply 24 volts plus or minus 0.5 volts ac, 400 cps between pins S and X of J201, the output voltages shall produce a null at input shaft and ratio settings of table I within the tolerance specified. Check for discontinuities in voltage output between 450 and 4700 meters by observing the oscilloscope display.

4.5.8 *Output voltages resistor R202 section 1.* To determine conformance to 3.6.4, the device used in 4.5.6 shall be attached to the input shaft. An ac ratio standard (4.4.2) shall be connected in parallel with resistor R202 section 1, by connection at pins small i and small k of J201. Connect a high impedance oscilloscope (4.4.2) between pin small m of J201 and the ratio standard slider. Apply 6.5 volts plus or minus 0.5 volts ac, 400 cps between pins small i and small k of J201. The output voltages shall produce a null at input shaft and ratio settings of table I within the tolerances specified. Check for discontinuities in voltage output between 450 and 4700 meters by observing the oscilloscope display.

4.5.8.1 *Output voltages resistor R202 section 2.* To determine conformance to 3.6.4, the device used in 4.5.6 shall be attached to the input shaft. An ac ratio standard (4.4.2) shall be connected in parallel with resistor R202, section 2, by connection at pins small t and small q of J201. Connect a high impedance oscilloscope (4.4.2) between pin small g of J201 and the ratio standard slider. Apply 6.5 volts plus or minus 0.5 volts ac, 400 cps between pins small t and small q of J201. The output voltages shall produce a null at input shaft and ratio setting of table I within the tolerances specified. Check for discontinuities in voltage output between 450 and 4700 meters by observing the oscilloscope display.

4.5.8.2 *Ouptut voltages resistor R202 section 3.* To determine conformance to 3.6.4, the device used in 4.5.6 shall be attached to the input shaft. An ac ratio standard (4.4.2) shall be connected in parallel with resistor R202, section 3, by connection at pins small f and small e of J201. Connect a high impedance oscilloscope (4.4.2) between pin large R of J201 and the ratio

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standard slider. Apply 6.5 volts plus or minus 0.5 volts ac, 400 cps between pins small f and small e of J201. The output voltages shall produce a null at input shaft and ratio setting of table I within the tolerances specified. Check for discontinuities in voltage output between 450 and 4700 meters by observing the oscilloscope display.

4.5.9 *Illumination.* To determine conformance to 3.7, the unit shall be connected to a 24 volt dc power source (4.4.2) and the range dial viewed in a darkened area. With power on, the dial shall be visible and legible.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marketing. Packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and the applicable packaging data sheet. The level of protection shall be as specified in the procurement document (see 6.2).

6. NOTES

6.1 Intended use. The input unit is primarily intended as a functional component of the electronic analog Computer, Ballistics: M16. The input unit is designed to receive mechanical rotation representative of the log of range from a range finder and convert this mechanical rotation to electrical voltage used as correction functions in the computer system.

6.2 Ordering data. Procurement documents should specify the following:
(a) Title, number and date of this specification.
(b) Selection of an applicable level of preservation packaging and packing (see 5.1).
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.

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